



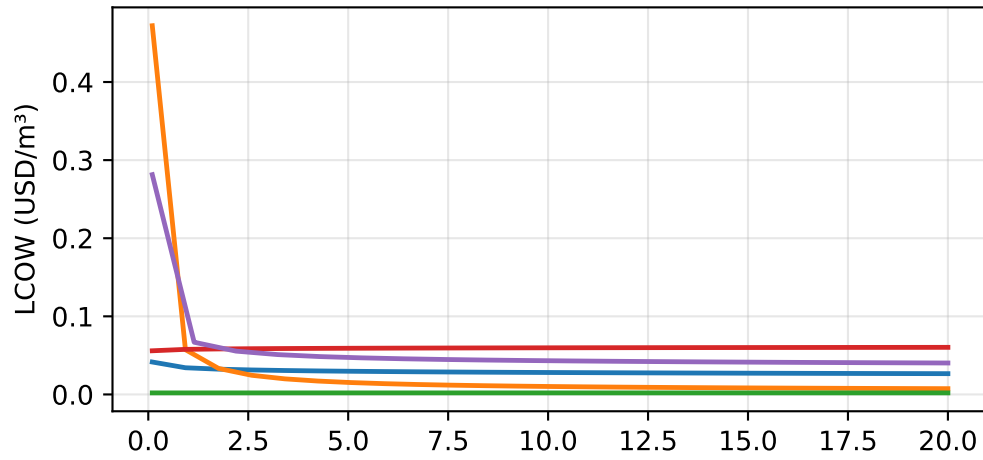
WaterTAP Zero-Order Model LCOW & CAPEX Curves

All costs in 2023 USD

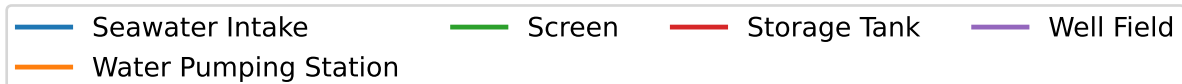
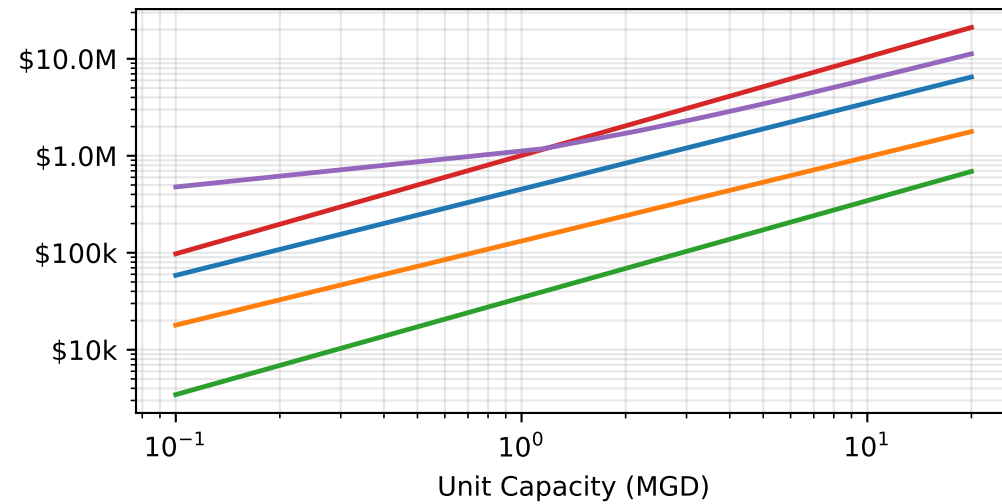
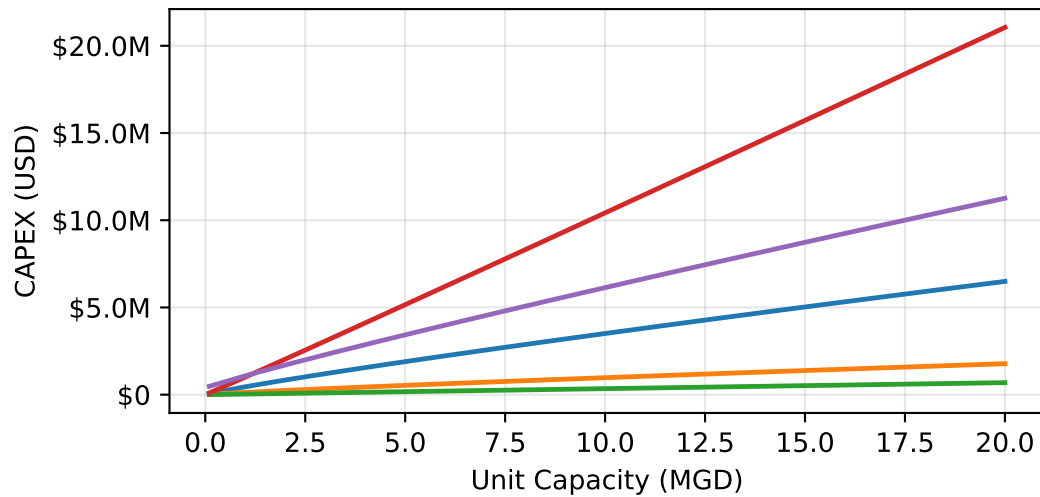
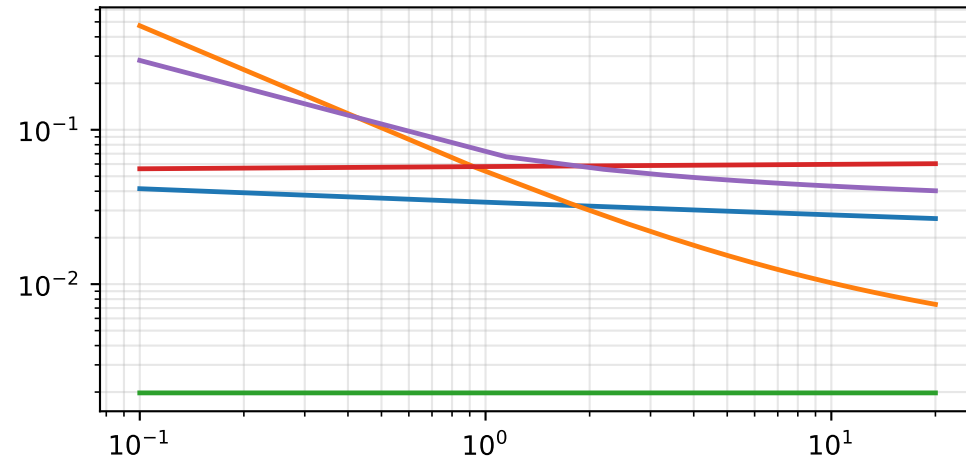
Generated from model runs

Intake & Conveyance

Linear Scale



Log-Log Scale

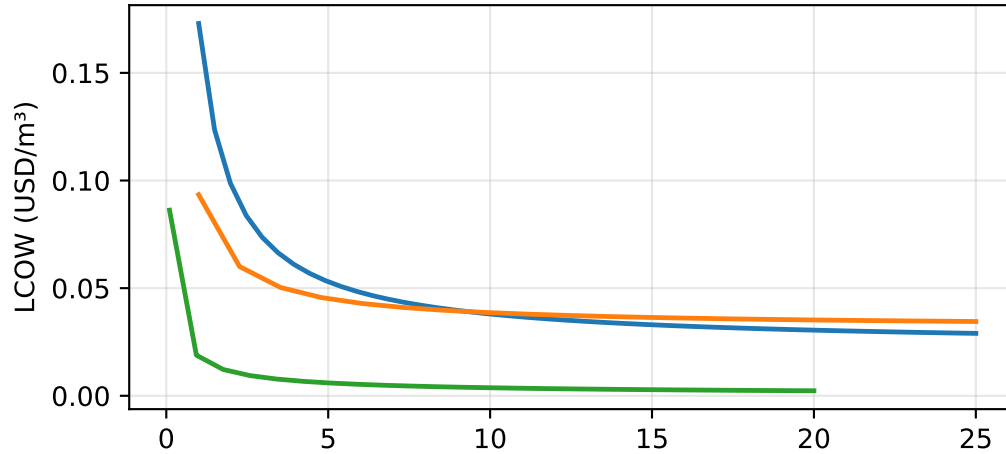


Details:
Well field results assume 1 mile of pipe distance.

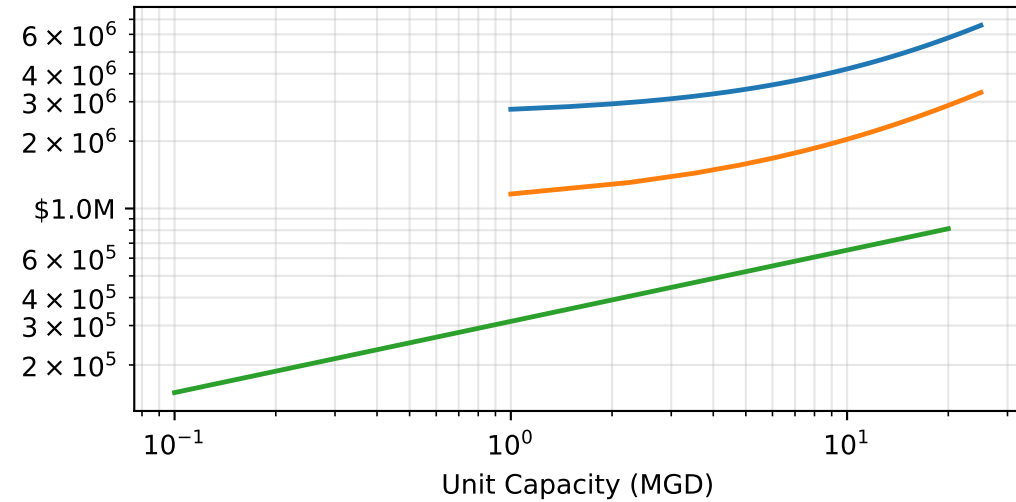
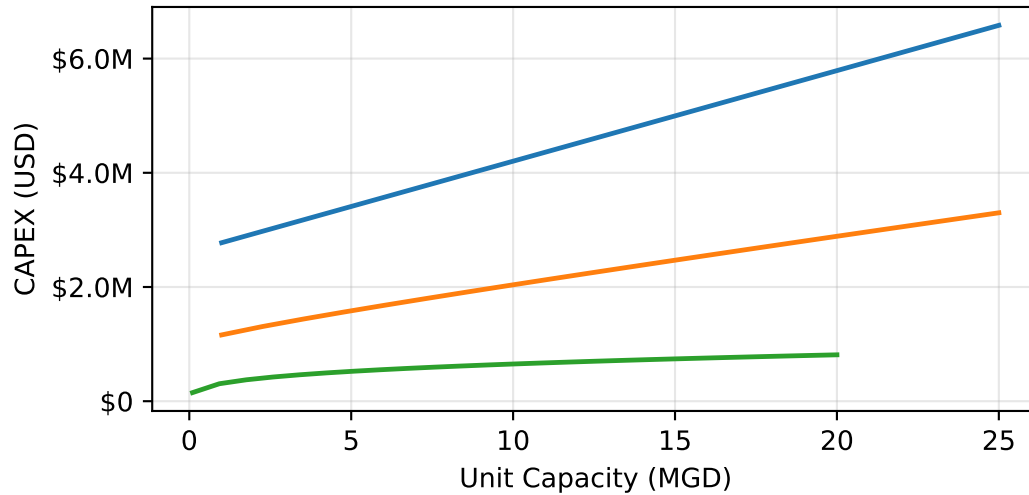
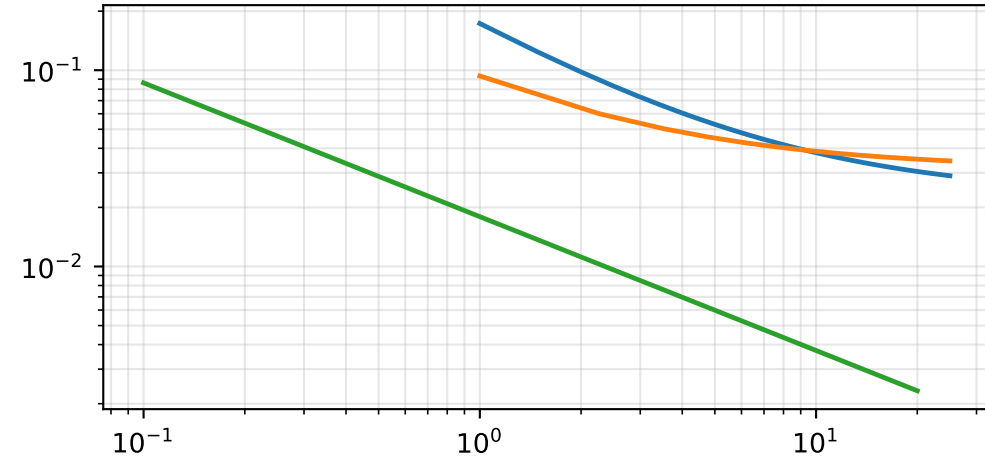


Pretreatment

Linear Scale



Log-Log Scale



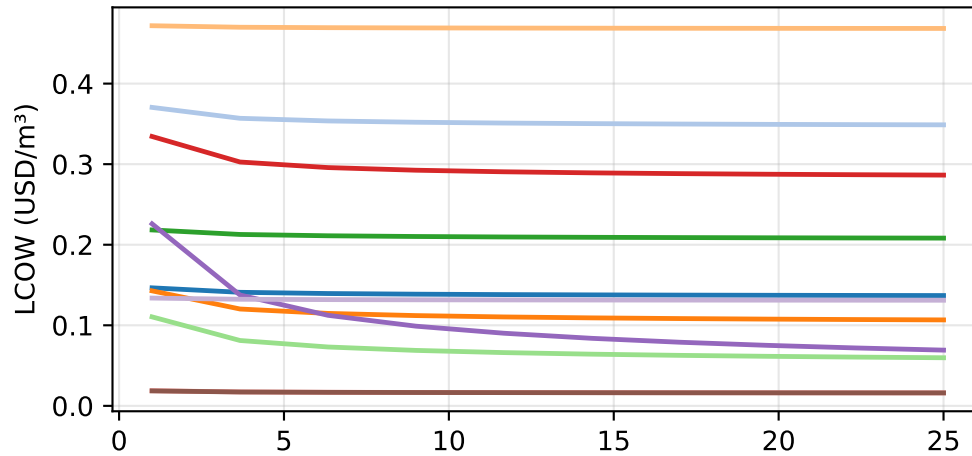
— Coagulation & Flocculation — Electrocoagulation — Sedimentation

Details:

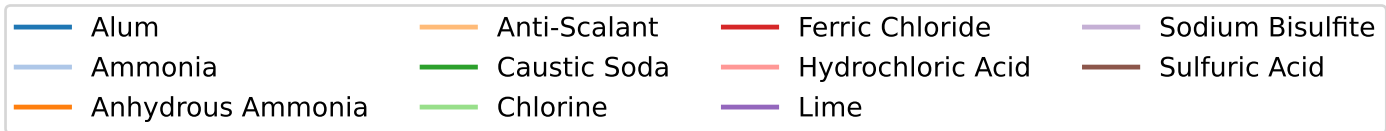
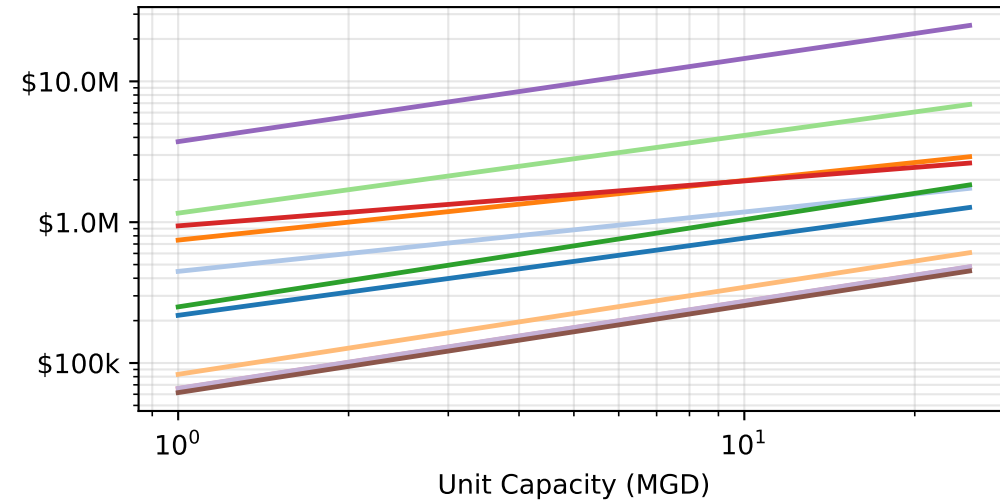
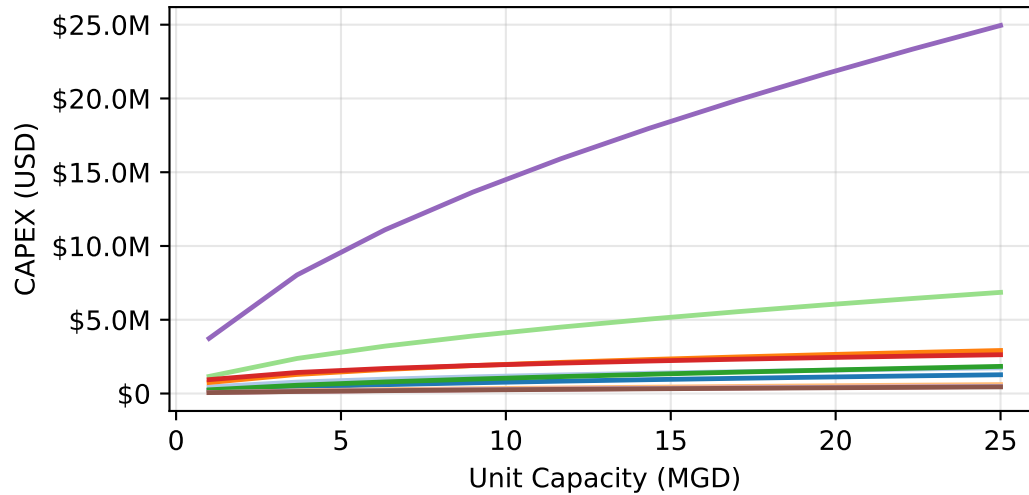
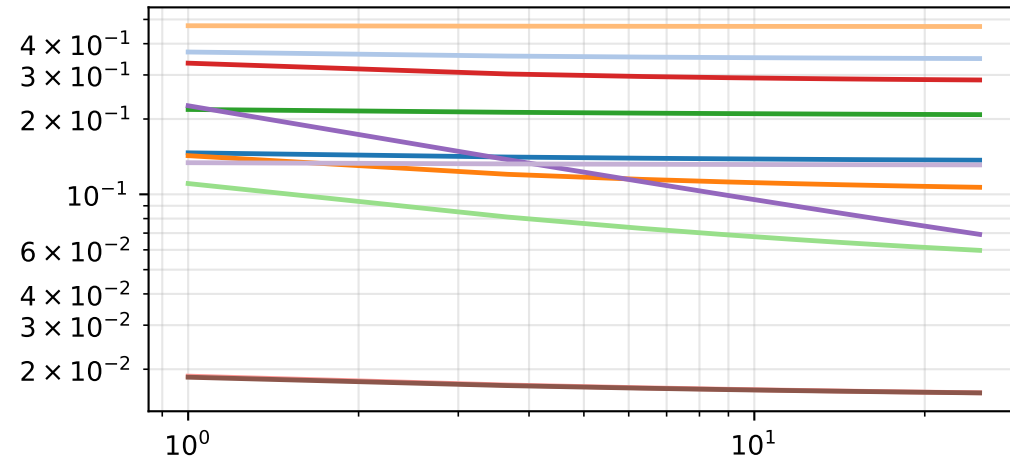
Electrocoagulation assumes a TDS of 1000 mg/L, a metal dose of 10 mg/L, and uses aluminum as the electrode material. Coagulation and flocculation are assumed to use 10 mg/L of alum and 0.1 mg/L of polymer.

Chemical Addition

Linear Scale



Log-Log Scale

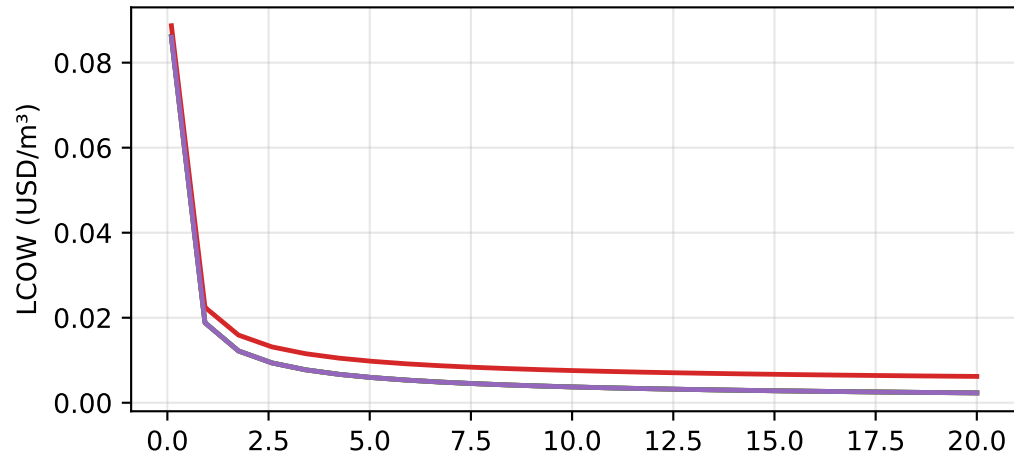


Details:
Chemical addition costs are run with a fixed dose of 100 mg/L for each chemical

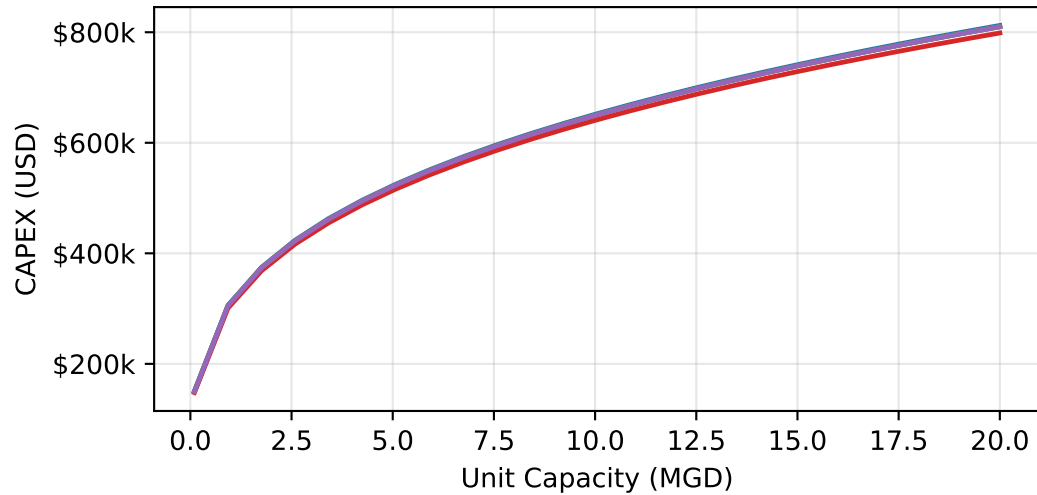
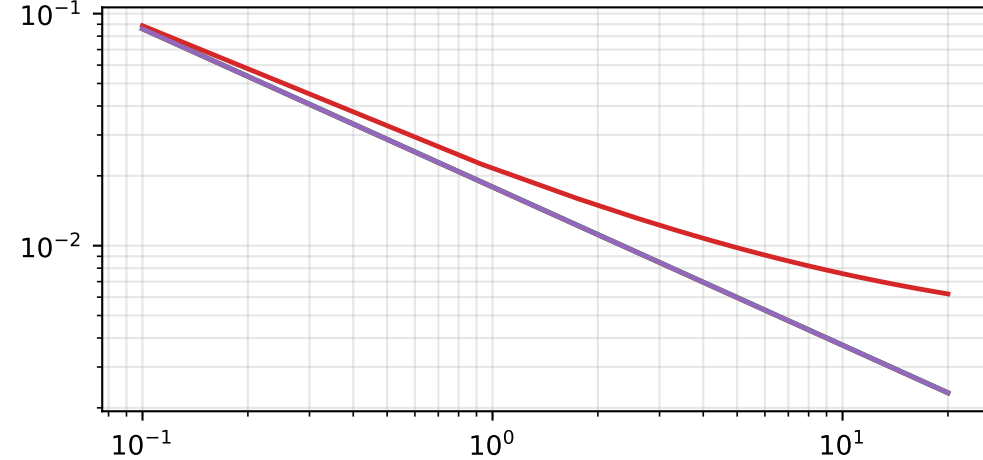


Clarification & Settling

Linear Scale

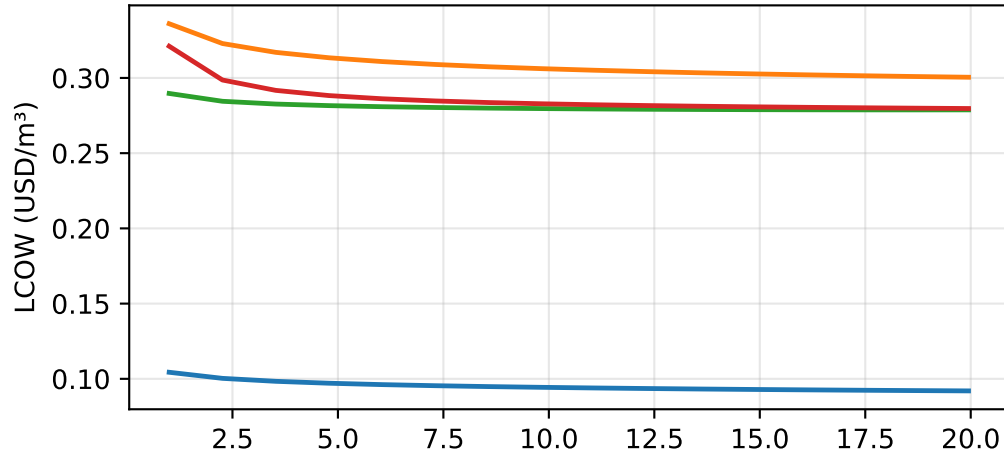


Log-Log Scale

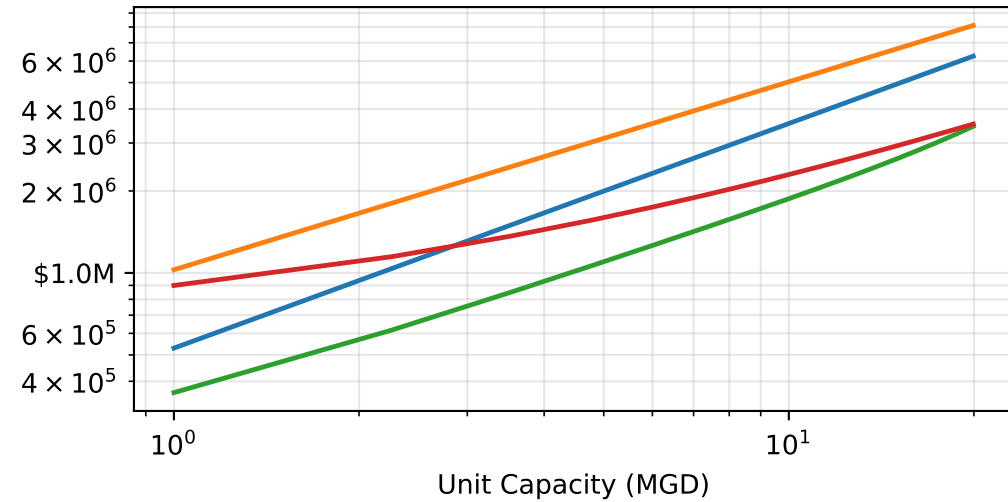
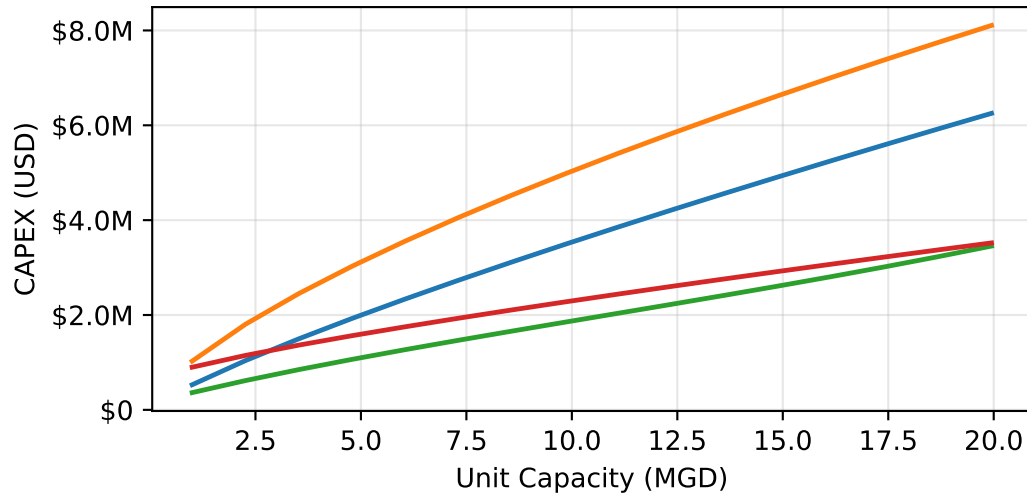
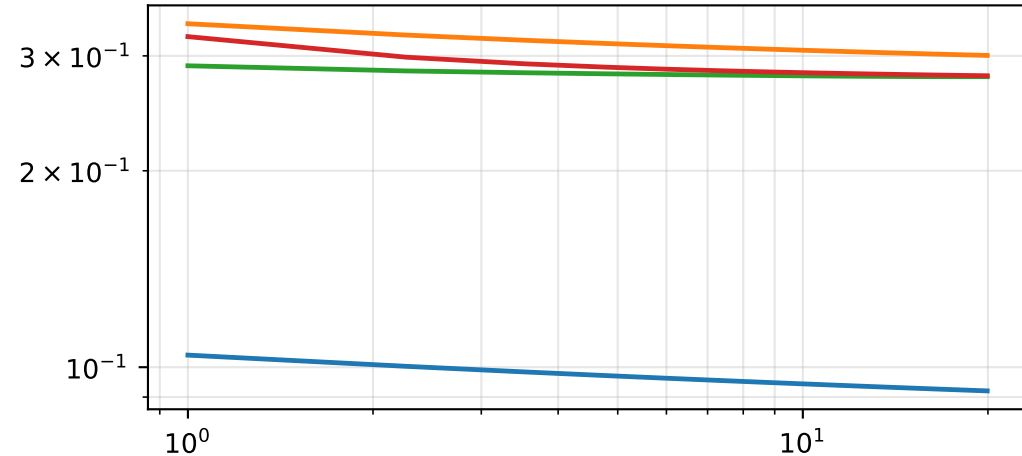


GAC & Ion Exchange

Linear Scale



Log-Log Scale



— Anion Exchange — Cation Exchange — GAC - Pressure Vessels — GAC - Gravity Basins

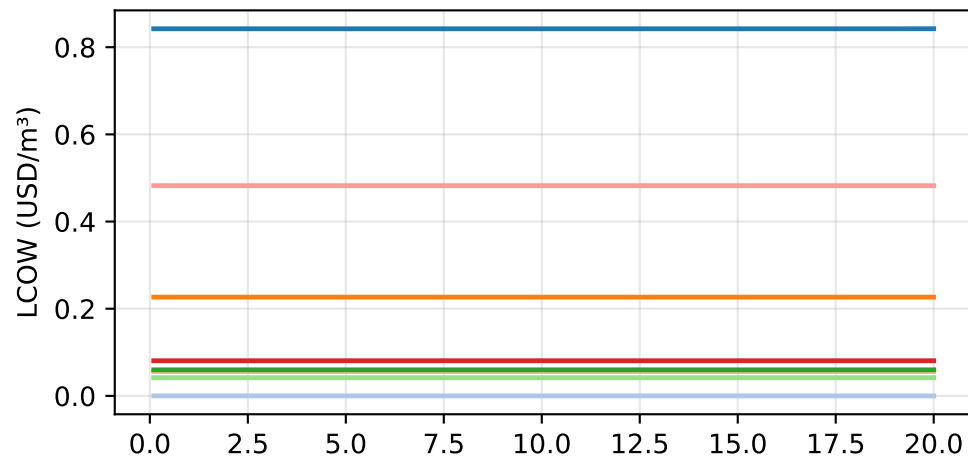
Details:

Anion exchange assumes 50 mg/L influent SO₄ concentration.
 Cation exchange assumes 200 mg/L inlet hardness concentration.
 GAC assumes 10 minute EBCT.

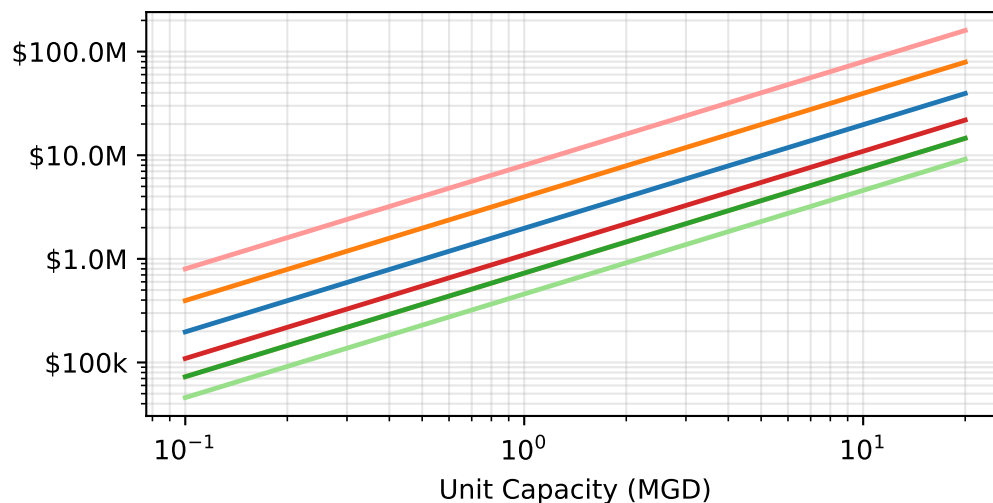
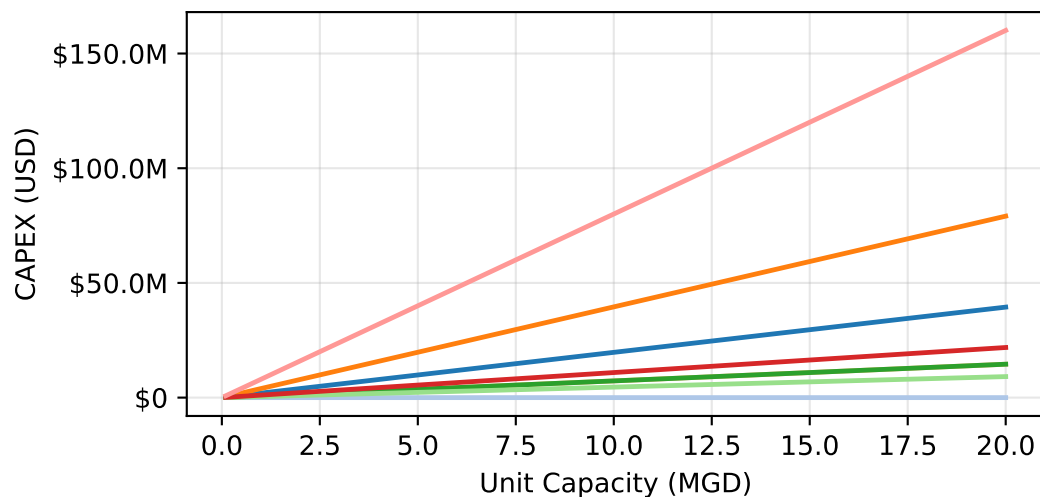
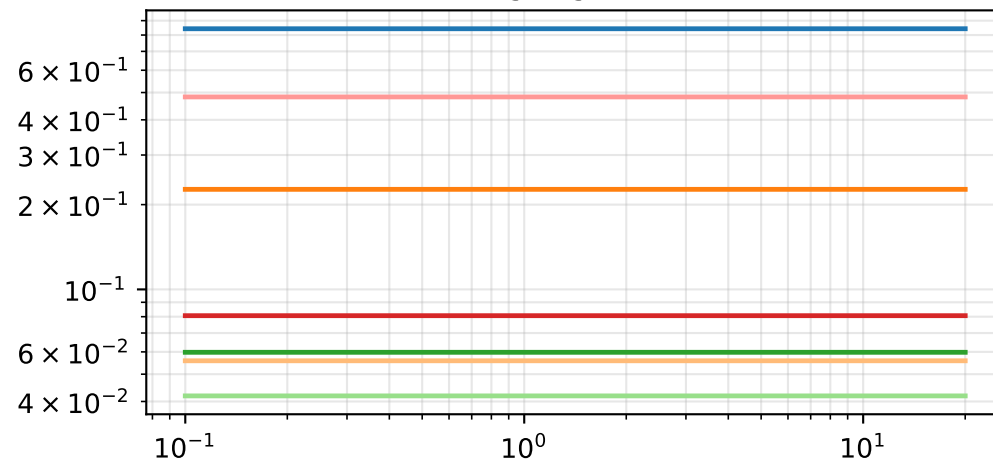


Membrane Processes

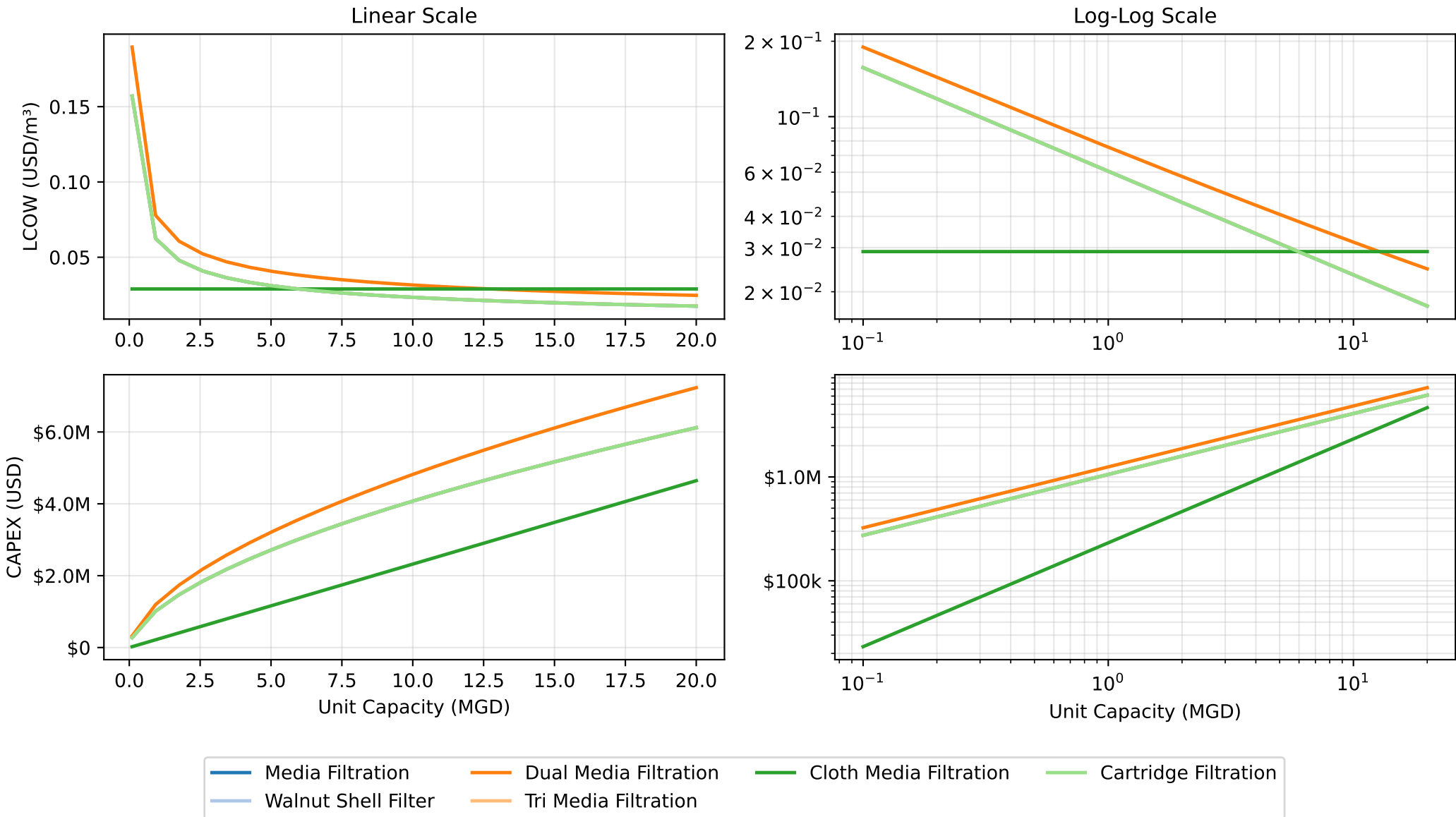
Linear Scale



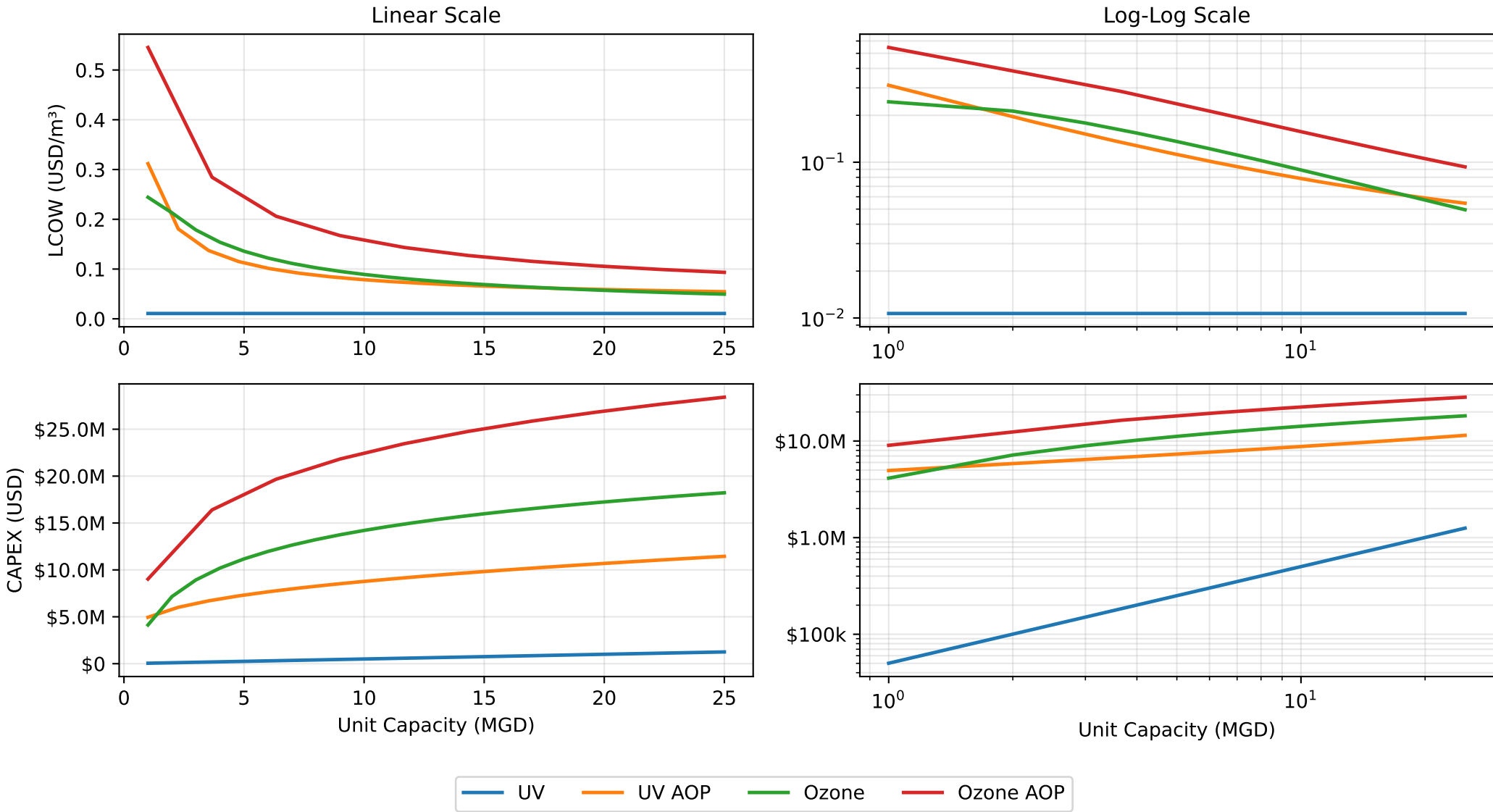
Log-Log Scale



Filtration Technologies



UV & Ozone AOP

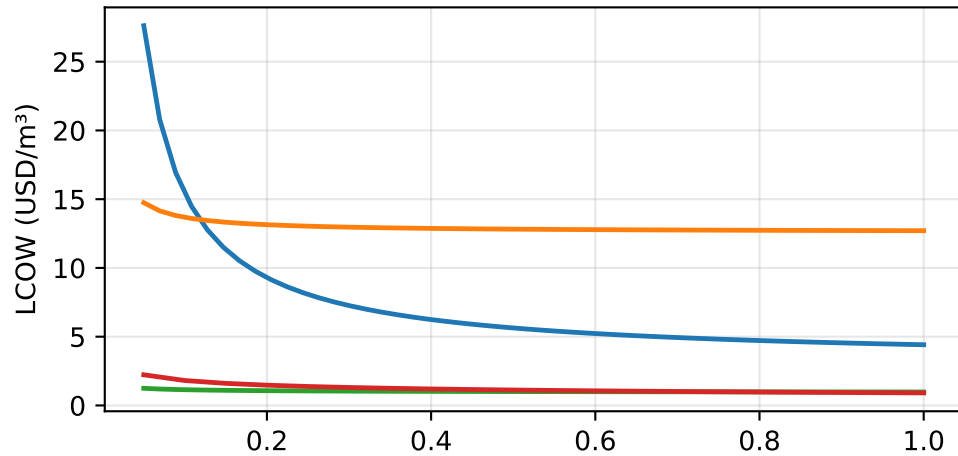


Details:

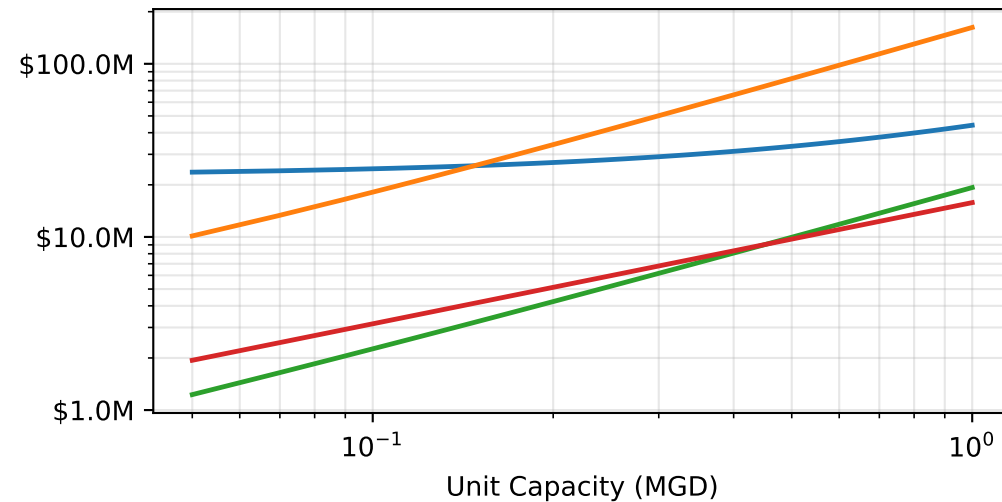
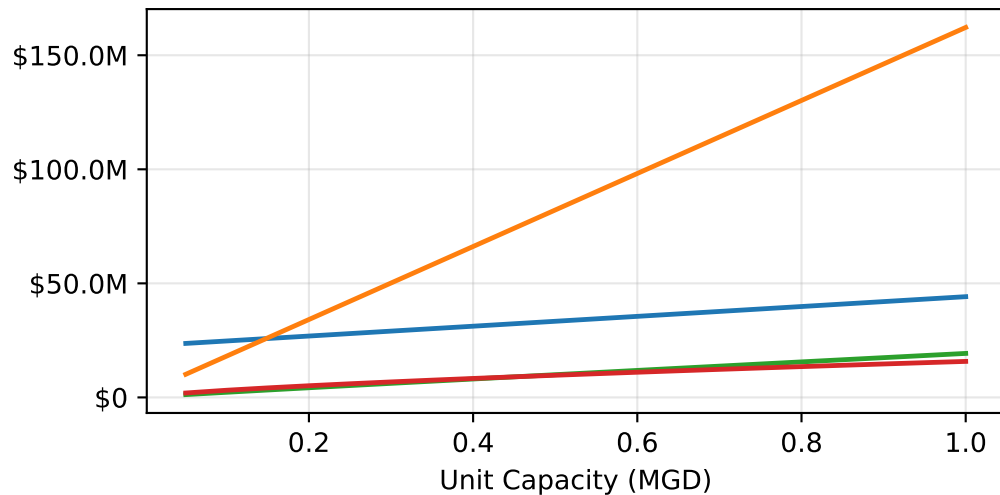
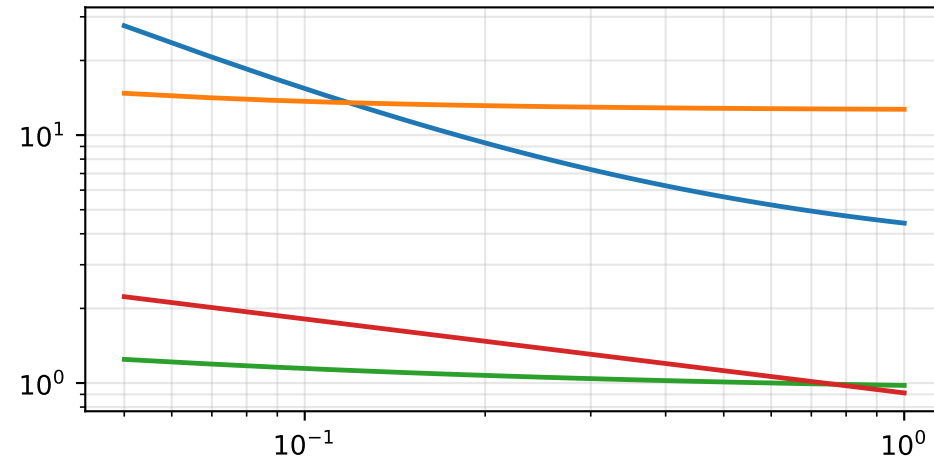
UV and ozone AOP costs are run with a fixed oxidant dose of 10 mg/L and UV SEC of 0.1 kWh/m³. Ozone costs are for a fixed ozone dose of 9 mg/L.

Brine Management

Linear Scale



Log-Log Scale



— Brine Concentrator — Crystallizer — Evaporation Pond — Deep Well Injection

Details:

Brine concentrator and crystallizer CAPEX and SEC are a function inlet flow rate and salinity.

These results are for a feed salinity of 60,000 mg/L TDS for the brine concentrator and 145,000 mg/L TDS for the crystallizer.

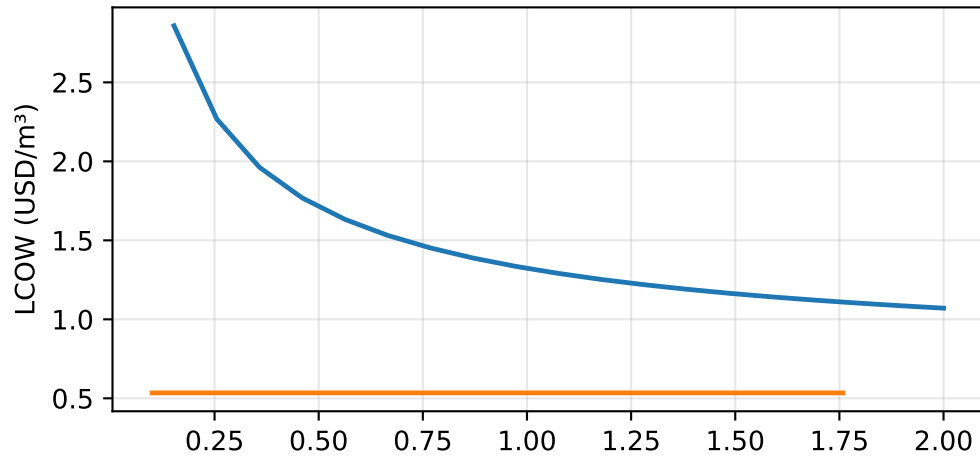
Evaporation pond costs are a function of inlet flow rate and solar radiation, with results shown for 25 MJ/m²/day.

Deep well injection costs are a function of inlet flow rate and pipe distance, with results shown for a pipe distance of 25 miles.

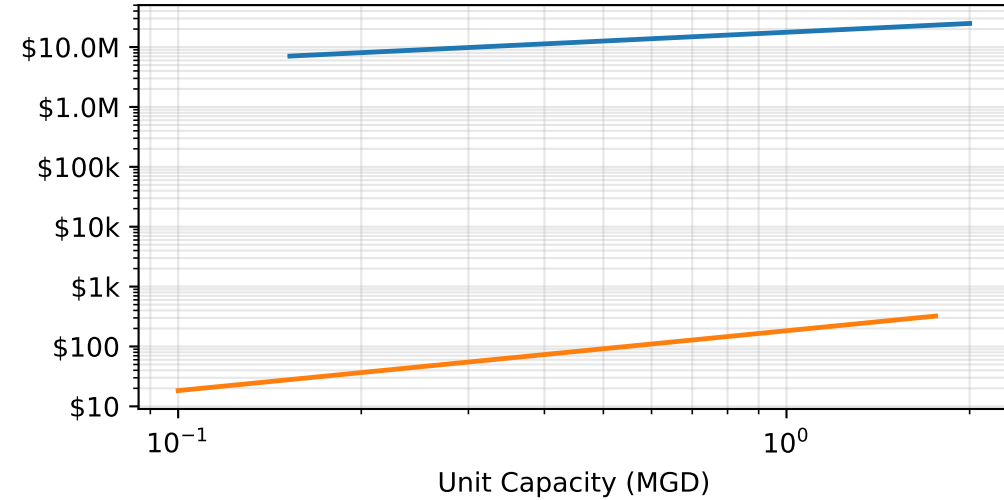
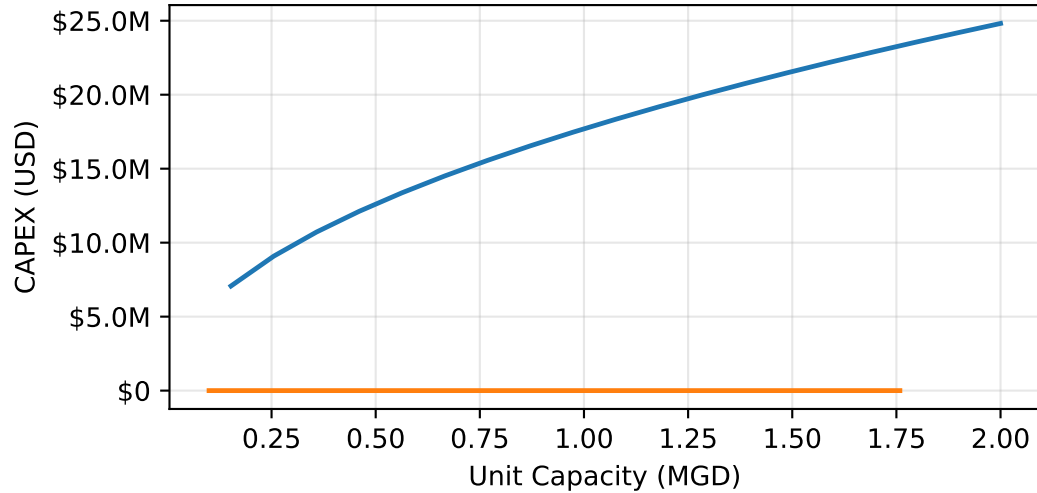
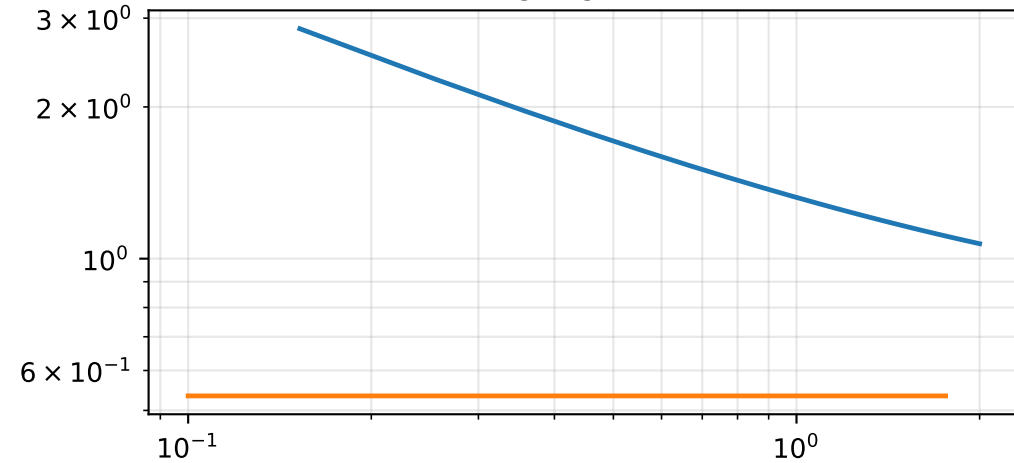


Solids Handling

Linear Scale



Log-Log Scale



Filter Press Centrifuge

WaterTAP Costing Reference Guide

Detailed Unit Model Costing Parameters

These models compute costs from physics-derived sizing variables. Capital cost equations vary by unit type. All parameters are tunable.

Unit Model	Capital Cost Eq.	Key Default Parameters	Reference
Anaerobic Digester	$A \times (Q/Q_{ref})^B$	A = \$19.36M B = 0.6 $Q_{ref} = 911 \text{ m}^3/\text{day}$	NREL Waste-to-Energy Model
CSTR	$A \times V^B$	A = \$1,246.1/m ³ B = 0.71	C.C. Tang, 1984
Chiller	$C_{chill} \times (W_{duty}/COP)$	$C_{chill} = \$200/\text{kW}$ COP = 7	
Clarifier (Circular)	$A \times A_{surf}^2 + B \times A_{surf} + C$	A = -6×10^{-4} B = 98.952 C = 191,806	Sharma et al. 2013
Clarifier (Primary)	$A \times Q_{MGD}^B$	A = \$538,746 B = 0.7	Byun et al. 2022
Clarifier (Rectangular)	$A \times A_{surf}^2 + B \times A_{surf} + C$	A = -2.9×10^{-3} B = 169.19 C = \$94,365	Sharma et al. 2013
Compressor	$C_{comp} \times W_{mech}^B$	$C_{comp} = \$7364$ B = 0.7	El-Sayed et al., 2001
Crystallizer	$A \times (Q/Q_{ref})^B \times IEC$	A = \$675,000 $Q_{ref} = 1 \text{ m}^3/\text{hr}$ B = 0.53 IEC = 1.43 steam = \$0.004/kg	Woods, 2007; Diab and Gerogiorgis, 2017; Yusuf et al., 2019; Panagopoulos, 2019
Dewatering (Centrifuge)	$A \times Q + B$	A = \$328.03/(gal/hr) B = \$751,295	McGivney & Kawamura, 2008
Dewatering (Filter Belt Press)	$A \times Q + B$	A = \$146.29/(gal/hr) B = \$433,972	McGivney & Kawamura, 2008
Dewatering (Filter Plate Press)	$A \times Q^B$	A = \$102,784/(gal/hr) B = 0.4216	McGivney & Kawamura, 2008
Electrocoagulation	$C_{electrodes} + C_{powersupply} + C_{reactor}$	$C_{electrodes} = \$2.23/\text{kg (aluminum)}, \$3.41/\text{kg (iron)}$ $C_{powersup} = \text{linear f(watt)}$ $C_{reactor} = \text{power f(vol)}$	McGivney & Kawamura, 2008; Smith, 2005; Anuf et al., 2022; magna-power.com
Electrodialysis	$C_{mem} \times A_{mem} + C_{electrode}$	$C_{mem} = \$160/\text{m}^2$ $C_{electrode} = \$2100/\text{m}^2$ replacement = 20 %/yr (mem + electrode)	Bian et al., 2018
Electrolyzer	$C_{mem} + C_{anode} + C_{cathode}$	$C_{mem} = \$25/\text{m}^2$ $C_{anode} = \$300/\text{m}^2$ $C_{cathode} = \$600/\text{m}^2$ mat fraction = 65 %	Bommaraju & O'Brien, 2015; Kent (Ed.), 2007; O'Brien, Bommaraju, & Hine, 2005; Yee, 2012
Energy Recovery Device	$C_{ERD} \times Q$	$C_{ERD} = \$535/(\text{m}^3/\text{s})$	
Evaporator	$C_{evap} \times A_{evap}$	$C_{evap} = \$1000/\text{m}^2$ material factor = 1.0	

GAC	$C_{\text{contactor}} + C_{\text{gac}} + C_{\text{other}}$	$C_{\text{contactor}}$ = polynomial f(vol) C_{gac} = exponential f(mass) C_{other} = power f(vol) C_{regen} = \$4.28/kg C_{makeup} = \$4.58/kg	EPA-WBS 2021
Heat Exchanger	$C_{\text{HX}} \times A_{\text{hx}}$	C_{HX} = \$300/m ² steam = \$0.008/kg	
Heater (Electric)	$C_{\text{heat}} \times W_{\text{duty}}/\text{eff}$	C_{heat} = \$66/kW eff = 0.99	
Ion Exchange (Cation / Anion)	$C_{\text{resin}} + C_{\text{vessels}} + C_{\text{tanks}}$ $C_i = A^b$	$C_{\text{res,AX}}$ = \$205/ft ³ $C_{\text{res,CX}}$ = \$153/ft ³ vessel A=1596.5, b=0.46 resin replacement = 5 %/yr	EPA-WBS 2021
Membrane Distillation	$C_{\text{mem}} \times A_{\text{mem}}$	C_{mem} = \$56/m ² replacement = 20 %/yr	Shamlou, Vidic, & Khanna, 2022
Mixer	$C_{\text{mix}} \times Q$	Generic = \$361/(L/s) NaOCl mixer = \$5.08/(m ³ /day) CaOH ₂ = 873.9/(kg/day)	
Nanofiltration	$C_{\text{mem}} \times A_{\text{mem}}$	C_{mem} = \$15/m ² replacement = 20 %/yr	
OARO	$C_{\text{mem}} \times A_{\text{mem}}$	C_{mem} = \$30/m ² $C_{\text{mem,HP}}$ = \$50/m ² replacement = 15 %/yr	Bartholomew et al. 2017
Pressure Exchanger	$C_{\text{PX}} \times Q$	C_{PX} = \$535/(m ³ /s)	
Pump (high pressure)	$C_{\text{pump}} \times W_{\text{mech}}$	C_{pump} = \$53/W	Malek et al. 1996
Pump (low pressure)	$C_{\text{pump}} \times Q$	C_{pump} = \$889/(m ³ /s)	Bartholomew et al. 2018
Reverse Osmosis	$C_{\text{mem}} \times A_{\text{mem}}$	C_{mem} = \$30/m ² $C_{\text{mem,HP}}$ = \$75/m ² replacement = 20 %/yr	Bartholomew et al. 2018
Steam Ejector	$A \times (M)^B$	M = motive steam + entrained vapor; kg/hr A = \$1949 B = 0.3 steam = \$0.008/kg	Gabriel, 2015
Thickener	$A \times A_{\text{surf}} + B$	A = 4729.8/ft ² B = \$37,068	McGivney & Kawamura, 2008
UV+AOP	$C_{\text{reactor}} + C_{\text{lamp}}$	C_{reactor} = \$202.35/kW C_{lamp} = \$235.50/kW lamp replacement = 33.3 %/yr	UV System Cost Analysis Tool (UVCAT); Wright, Gaithuma, Greene, Aieta, 2006

WaterTAP Costing Reference Guide (continued)

Zero-Order Unit Model Costing — Default Parameters

Units with A and B values follow the equation: $C_{\text{cap}} = A \times (Q_{\text{in}}/Q_{\text{basis}})^B$. A is in USD for the reference year shown.

Unit Model	A	B	Q _{basis}	Cost Year	Energy	Recovery	Validity Range	Reference
Aeration Basin	725,570.0	0.586	1 MGD	2014	0.412	N/A	0-30 MGD	Texas Water Development Board IT3PR version 2.0, Figure 3.3
Anaerobic Digestion Oxidation	19,355,231.2	0.600	911 m³/hr	2012	0.15	0.9999	—	NREL Waste-to-Energy Model (WESyS); WT3 Beef Processing case study
Backwash Solids Handling	41,812.0	0.918	1 MGD	2007	$f(Q)^S$	0.95	—	Texas Water Development Board IT3PR version 2.0 User Manual; Based on costs for subprocesses in Table 5.7.1 (Equations 29, 30, 31, 32, 33, 37, 38, 39) in Cost Estimating Manual for Water Treatment Facilities (McGivney/Kawamura); Weighted averages of exponents for equations
Bio-Active Filtration	725,570.0	0.586	1 MGD	2014	None	0.99	0-30 MGD	Defined based on the San Luis case study data.; Texas Water Development Board IT3PR version 2.0, Figure 3.3
Bioreactor	1,252,160.0	1	45 m³/hr	2018	None	0.97	—	Defined based on the San Luis case study data.; Developed with Aspen Cost Estimation for San Luis Case Study in WT3.
Brine Concentrator	†	†	—	2007	$f(x)^S$	0.9	Q _{in} : 0-20 MGD RR: 56%-98% TDS: 4,000-61,851 mg/L	Survey of High-Recovery and Zero Liquid Discharge Technologies for Water Utilities (2008). Table 5.1, Table A2.3
Cartridge Filtration	725,570.0	0.586	1 MGD	2014	0.0002	0.9999	0-30 MGD	Texas Water Development Board IT3P Figure 3.3
Chemical Addition (Alum)	15,408.0	0.548	—	2007	$f(x)$	N/A	2-1,000 gal/hr	McGivney & Kawamura, Figure 5.5.6 + Table 5.2.1, Cost Estimating Manual for Water Treatment Facilities
Chemical Addition (Ammonia)	6,699.1	0.422	—	2007	$f(x)$	N/A	50-750 gal/day	McGivney & Kawamura, Figure 5.5.15, Cost Estimating Manual for Water Treatment Facilities
Chemical Addition (Anhydrous Ammonia)	7,958.7	0.423	—	2007	$f(x)$	N/A	240-5,080 lb/day	McGivney & Kawamura, Figure 5.5.14 + Table 5.2.1, Cost Estimating Manual for Water Treatment Facilities
Chemical Addition (Anti-Scalant)	901.0	0.618	—	2007	$f(x)$	N/A	11-5,300 gal/day	McGivney & Kawamura, Figure 5.5.11 + Table 5.2.1, Cost Estimating Manual for Water Treatment Facilities
Chemical Addition (Caustic Soda)	2,262.8	0.620	—	2007	$f(x)$	N/A	9-10,000 gal/day	McGivney & Kawamura, Figure 5.5.12b + Table 5.2.1, Cost Estimating Manual for Water Treatment Facilities
Chemical Addition (Chlorine)	5,207.4	0.552	—	2007	$f(x)$	N/A	200-10,000 lb/day	McGivney & Kawamura, Figure 5.5.1b + Table 5.2.1, Cost Estimating Manual for Water Treatment Facilities
Chemical Addition (Ferric Chloride)	34,153.0	0.319	—	2007	$f(x)$	N/A	2-1,450 gal/day	McGivney & Kawamura, Figure 5.5.13, Cost Estimating Manual for Water Treatment Facilities
Chemical Addition (Hydrochloric Acid)	901.0	0.618	—	2007	$f(x)$	N/A	11-5,300 gal/day	McGivney & Kawamura, Figure 5.5.11 + Table 5.2.1, Cost Estimating Manual for Water Treatment Facilities
Chemical Addition (Lime)	12,985.0	0.590	—	2007	$f(x)$	N/A	10-10,000 lb/day	McGivney & Kawamura, Figure 5.5.9 + Table 5.2.1, Cost Estimating Manual for Water Treatment Facilities
Chemical Addition (Sodium Bisulfite)	901.0	0.618	—	2007	$f(x)$	N/A	11-5,300 gal/day	McGivney & Kawamura, Figure 5.5.11 + Table 5.2.1, Cost Estimating Manual for Water Treatment Facilities

Chemical Addition (Sulfuric Acid)	901.0	0.618	—	2007	f(x)	N/A	11-5,300 gal/day	McGivney & Kawamura, Figure 5.5.11 + Table 5.2.1, Cost Estimating Manual for Water Treatment Facilities
Chlorination	†	†	—	2014	5e-05	N/A	Q_{in} : 1-25 MGD $[Cl_2]$: 1-25 mg/L	Texas Water Development Board IT3PR Table 3.23; Texas Water Development Board IT3PR, model default; EPA Guidance Manual Disinfection Profiling and Benchmarking from Table B-1, for temperature of 20 C and pH of 8, chlorine concentration of 1 mg/L
Clarifier	177,516.0	0.318	5,450 m³/hr	2007	None	0.999	A_{basin} : 5,000-150,000 ft²	McGivney & Kawamura, Figure 5.5.24c + Table 5.2.1, Cost Estimating Manual for Water Treatment Facilities
Coagulation & Flocculation	†	†	—	2007	f(x)	0.9	V_{mix} : 800-145,000 gal m_{Alum} : 10-5,070 lb/hr m_{poly} : 1-220 lb/day V_{floc} : 0-7 × 10⁶ gal	McGivney & Kawamura Figure 5.5.19, Figure 5.5.22, 5.5.7b, 5.5.8, Cost Estimating Manual for Water Treatment Facilities
Conventional Activated Sludge	725,570.0	0.586	1 MGD	2014	0.14	0.9999	0-30 MGD	Texas Water Development Board IT3PR version 2.0 User Manual; Table 17-3 in "Wastewater Engineering - Treatment and Reuse" by Metcalf & Eddy; Texas Water Development Board IT3PR version 2.0, Figure 3.3
Cooling Supply	40,299.0	0.866	1 MGD	2007	None	N/A	0-300 MGD	McGivney & Kawamura Figure 5.5.35b
Crystallizer	†	†	—	2007	f(x)	0.95	Q_{in} : 11-1,047 gpm f_{purge} : 0%-52% TDS: 145,000-358,000 mg/L	Survey of High-Recovery and Zero Liquid Discharge Technologies for Water Utilities (2008). Table 4.4, Table A2.1, Table A2.3; Default value based on 5% purge fraction per source. Survey of High-Recovery and Zero Liquid Discharge Technologies for Water Utilities (2008).
Decarbonator	150,000.0	1	1,577 m³/hr	2007	None	N/A	—	Survey of High-Recovery and Zero Liquid Discharge Technologies for Water Utilities (2008). Table A2.2.
Deep Well Injection	†	†	—	2011; 2020 [‡]	f(Q)	N/A	—	Produced water case studies; KBHDP case study
Dual Media Filtration	3,481,852.0	0.586	30 MGD	2007	0.0507	0.99	0-30 MGD	McGivney & Kawamura Figure 5.5.27 + 5.5.29, assumes 6 media filtration units and 1 filter backwash unit; Texas Water Development Board IT3PR version 2.0, Figure 3.3; None
Electrocoagulation	†	†	—	2000; 2007; 2020	f(x)	0.99	—	Agitated Reactor from Table 2.1 in Chemical Process - Design and Integration; Smith (2005) - doi.org/10.1205/cherd.br.0509; Z. Gu, Z. Liao, M. Schulz, J. R. Davis, J. C. Baygents and J. Farrell (2009); DOI- 10.1021/ie801086c; Uludag-Demirer et al., 2020 - doi.org/10.3390/su12072697; McGivney & Kawamura, Figure 5.5.22 + Table 5.2.1, Cost Estimating Manual for Water Treatment Facilities
Electrodialysis Reversal	31,000,000.0	1	946 m³/hr	2016	f(x)	0.9214	—	Based on costs from Fort Irwin / Irwin Water Works case study; Membrane Technology and Applications; R. W. Baker (2012) DOI- 10.1002/9781118359686. Fit to data from Figure 10.15
Evaporation Pond	†	†	—	2007	None	0.0001	10-100 acre	Survey of High-Recovery and Zero Liquid Discharge Technologies for Water Utilities (2008).; BLM, Membrane Concentrate Disposal - Practices and Regulation (2006), Section 10.6

Filter Press	61,672.1	0.488	—	2007	f(x)	0.0001	30-6,600 gal/hr	Regressed from data in Fig 9 Chap 18 of "Biosolids Treatment Processes" for 6% solids; doi 10.1007/978-1-59259-996-7; McGivney & Kawamura, Figure 5.5.40 + Table 5.2.1, Cost Estimating Manual for Water Treatment Facilities
Fixed Bed Bioreactor	1,542,008.0	0.734	1 MGD	2017	0.0309	N/A	1-75 MGD	Average from standard designs for EPA-WBS Biological Treatment with Fixed Bed using pressure vessels from 2017; Regressed from standard designs for EPA-WBS Biological Treatment with Fixed Bed using pressure vessels from 2017
GAC	†	†	—	2020	0.106	0.96	—	Heap leaching (gold/copper) case study.
Intrusion Mitigation	44,600,000.0	1	4,750 m³/hr	1998	0.0512	N/A	—	Monterey One Case Study Data
Ion Exchange (Anion Exchange)	†	†	—	—	0.00542	N/A	Q _{in} : 0-75 MGD [SO ₄]: 50-150 mg/L	Regressed from output from standard designs from EPA-WBS Anion Exchange model; Average from output from standard designs from EPA-WBS Anion Exchange model
Ion Exchange (Cation Exchange)	†	†	—	—	0.069	N/A	Q _{in} : 0-75 MGD TDS: 200-1,000 mg/L	Average from output from standard designs from EPA-WBS Cation Exchange model; Regressed from output from standard designs from EPA-WBS Cation Exchange model
Media Filtration	725,570.0	0.586	1 MGD	2014	0.00015	0.9999	0-30 MGD	Texas Water Development Board IT3PR version 2.0, Figure 3.3
Membrane Bioreactor	725,570.0	0.586	1 MGD	2014	f(Q)	0.9999	1-30 MGD	Texas Water Development Board IT3PR version 2.0, Figure 3.3; Assessing Location and Scale of Urban Nonpotable Water Reuse Systems for Life-Cycle Energy Consumption and Greenhouse Gas Emissions, Figure 3, Table S3, Eq. S.11 - Kavvada et al (2016)
Microfiltration	500,000.0	1	1 MGD	2014	0.18	0.95	—	Texas Water Development Board IT3PR Table 3.21
Municipal Drinking	40,299.0	0.866	1 MGD	2007	f(Q)	N/A	0-300 MGD	McGivney & Kawamura Figure 5.5.35b
Nanofiltration	750,000.0	1	1 MGD	2014	0.231	0.85	—	Based on Vogtle case study data.; Texas Water Development Board IT3P Table 3.21
Ozone	†	†	—	—	f(x)	N/A	Q _{in} : 1-25 MGD [O ₃]: 1-25 mg/L	Texas Water Development Board IT3PR version 2.0 User Manual; EPA Disinfection Profiling and Benchmarking - Appendix B, Table B-6 for 25 C; B. Mundy et al. (2018) doi.org/10.1080/01919512.2018.1467187 - Energy consumption for ozone generator, ozone destruct, nitrogen feed system, cooling water pumps, and other misc. energy uses.; Texas Water Development Board IT3P Table 3.24

Ozone+AOP	†	†	—	—	f(x)	N/A	Q _{in} : 1-25 MGD [O ₃]: 1-25 mg/L	EPA Disinfection Profiling and Benchmarking - Appendix B, Table B-6 for 25 C; B. Mundy et al. (2018) doi.org/10.1080/01919512.2018.1467187 - Energy consumption for ozone generator, ozone destruct, nitrogen feed system, cooling water pumps, and other misc. energy uses.; Texas Water Development Board IT3PR User Manual Table 3.24; Texas Water Development Board IT3PR User Manual; Texas Water Development Board IT3PR User Manual Figure 3.4
Primary Separator	177,516.0	0.318	5,450 m³/hr	2007	None	0.999	A _{basin} : 0-160,000 ft²	Defined based on the Iron and Steel case study data.; McGivney & Kawamura, Figure 5.5.24c, Cost Estimating Manual for Water Treatment Facilities
Pump	40,299.0	0.866	1 MGD	2007	0.051	N/A	0-300 MGD	McGivney & Kawamura Figure 5.5.35b
Screen (Band)	10.3	1	1 m³/d	2018	None	0.9999	10,000-400,000 m³/d	Voutchkov, N. (2018). Desalination Project Cost Estimating and Management. Figure 4.5
Screen (Drum)	13.0	1	1 m³/d	2018	None	0.9999	10,000-400,000 m³/d	Voutchkov, N. (2018). Desalination Project Cost Estimating and Management. Figure 4.5
Screen (Micro)	8.9	1	1 m³/d	2018	None	0.99999	10,000-400,000 m³/d	Voutchkov, N. (2018). Desalination Project Cost Estimating and Management. Figure 4.7
Seawater Intake	3,624.0	0.889	1 m³/hr	2018	f(Q)	N/A	0-400,000 m³/d	Voutchkov, N. (2018). Desalination Project Cost Estimating and Management, Figure 4.2 + Figure 4.4
Sedimentation	13,572.0	0.318	—	2007	None	0.9999	A _{basin} : 5,000-150,000 ft²	Water and Wastewater Engineering - Design Principles and Practice (Mackenzie L. Davis); McGivney & Kawamura, Figure 5.5.24c + Table 5.2.1, Cost Estimating Manual for Water Treatment Facilities
Settling Pond	177,516.0	0.318	5,450 m³/hr	2007	None	0.9999	A _{basin} : 5,000-150,000 ft²	McGivney & Kawamura, Figure 5.5.24c + Table 5.2.1, Cost Estimating Manual for Water Treatment Facilities
Static Mixer	1,698.1	0.325	1 L/s	2010	None	N/A	1-50 L/s	Chemical Engineering Design, 2nd Edition. Principles, Practice and Economics of Plant and Process Design. Table 7.2 eq fit to power eq.
Surface Discharge	647.8	0.876	1 m³/d	2020	f(Q)	N/A	0-200,000 m³/d	Produced water case studies
Tri Media Filtration	725,570.0	0.586	1 MGD	2014	0.00045	0.9	0-30 MGD	Texas Water Development Board IT3PR version 2.0 User Manual; Texas Water Development Board IT3PR version 2.0, Figure 3.3
UV	†	†	—	2020	0.1	N/A	—	Developed from data in UV System Cost Analysis Tool (UVCAT), prepared by Carollo for NYSERDA
UV+AOP	†	†	—	2014; 2020	0.1	N/A	—	WT3 OCWD case study; Developed from data in UV System Cost Analysis Tool (UVCAT), prepared by Carollo for NYSERDA; Texas Water Development Board IT3PR User Manual Figure 3.4
Ultrafiltration	500,000.0	1	1 MGD	2014	0.231	0.95	—	Texas Water Development Board IT3PR version 2.0 User Manual; Texas Water Development Board IT3P Table 3.21
Walnut Shell Filter	725,570.0	0.586	1 MGD	2014	None	0.99	0-30 MGD	Defined based on the Kern County case study data.; Texas Water Development Board IT3PR version 2.0, Figure 3.3

Water Pumping Station (Raw)	24,348.0	0.867	1 MGD	2007	f(Q)	N/A	0-200 MGD	McGivney & Kawamura Figure 5.5.36b
Water Pumping Station (Treated)	40,299.0	0.866	1 MGD	2007	f(Q)	N/A	0-300 MGD	McGivney & Kawamura Figure 5.5.35b
Well Field	4,731.6	0.920	1 m³/hr	2018	f(Q)	N/A	1,000-30,000 m³/d	Voutchkov, N. (2018). Desalination Project Cost Estimating and Management, Table 4.7; 50 m deep wells; Produced water case studies

MGD = million gallons per day; m³ = cubic meters; ft² = square feet; RR = water recovery ratio; f_{purge} = purge fraction; m_x = mass flow rate of x; TDS = total dissolved solids; Q = volumetric flow rate

†: units use an alternate costing method

§: f(x) indicates energy calculated from other variables, f(Q) indicates energy calculated with alternative relationship as function of influent flow rate.

‡: If multiple Cost Year values, costing method includes parameters from several years.

Chemical addition units costed with the volumetric or mass-based flow of the chemical.